

THE ROLE OF DIGITAL BANKING FEATURES IN BANK SELECTION AN ANALYSIS OF CUSTOMER PREFERENCES FOR ONLINE AND MOBILE BANKING

Nur Al Faisal ¹

¹Assistant Professor of Finance, Texas A&M International University, Texas, USA
Corresponding Email: nur.faisal@tamiu.edu

Janifer Nahar ²

²Graduate Research Assistant, Department of Finance, Louisiana State University, Baton Rouge, Louisiana, USA
Email: janifernahar@gmail.com

Md. Waliullah ³

³Master of Science in Management Information Systems, College of Business, Lamar University, Texas, USA
Email: mwaliullah@lamar.edu

Razia Sultana Borna ⁴

⁴Master of Science in Management Information Systems, College of Business, Lamar University, Texas, USA
Email: ronjitaborna07@gmail.com

Keywords

Digital Banking
Bank Selection
Customer Preferences
Online Banking
Mobile Banking
Customer Satisfaction

ABSTRACT

This study explores the critical factors influencing the adoption and effectiveness of digital banking, focusing on customer preferences, technological advancements, and external pressures such as market competition and the COVID-19 pandemic. By conducting a systematic literature review of 112 peer-reviewed articles, this research examines key themes, including convenience, security, personalization, competitive innovation, and resilience during crises. The findings reveal that user-friendly interfaces, 24/7 availability, and seamless transaction processing are primary drivers of customer satisfaction and loyalty. Robust security measures, such as encryption and multi-factor authentication, and transparency in data usage policies play pivotal roles in fostering trust. Personalization, enabled by AI and data analytics, emerged as a significant factor in enhancing user experience and retention, though privacy concerns remain a challenge. Furthermore, the review highlights how market competition drives technological innovation, with traditional banks adopting blockchain, AI, and biometrics to maintain a competitive edge. The pandemic served as a catalyst for digital adoption, compelling banks to rapidly enhance their digital infrastructure to meet evolving customer needs. By synthesizing these insights, this study not only validates existing research but also identifies gaps and areas for future exploration, such as the ethical use of data and inclusivity in digital banking. These findings offer actionable insights for financial institutions and policymakers to optimize digital banking ecosystems for long-term growth and customer engagement.

1 INTRODUCTION

The evolution of digital banking has marked a paradigm shift in the financial services sector, driven by the

proliferation of technological advancements and the growing demand for seamless financial interactions (Chauhan et al., 2022). Digital banking, encompassing online and mobile banking services, has redefined

customer expectations and enabled financial institutions to offer more personalized and efficient services (Larsson & Viitaoja, 2017). These platforms facilitate round-the-clock access to banking services, empowering customers to perform transactions, monitor accounts, and even apply for loans from the convenience of their homes (Chauhan et al., 2022). As financial institutions worldwide adapt to the digital transformation, understanding the features that influence customer preferences has become imperative for sustaining competitiveness in the digital era. Moreover, Customer preferences for online and mobile banking features vary significantly based on factors such as ease of use, security, and service accessibility (Mbama & Ezepeue, 2018). The Technology Acceptance Model (TAM) underscores the importance of perceived usefulness and ease of use in fostering customer adoption of new technologies (Chauhan et al., 2021). Similarly, studies have found that the availability of user-friendly interfaces, real-time notifications, and secure login protocols significantly influence customer satisfaction and loyalty (Boon-itt, 2015). With the increasing integration of artificial intelligence and data analytics, customers now demand personalized experiences, making customization a pivotal aspect of digital banking platforms (Gigante et al., 2022).

In addition, Security and privacy concerns remain critical factors influencing customer preferences for digital banking. Despite technological advancements, cyber threats and data breaches have heightened customers' apprehensions about the safety of their financial information (Chauhan et al., 2022). Customers

prioritize banks that offer robust encryption mechanisms, multi-factor authentication, and fraud detection systems to ensure secure transactions (Mbama & Ezepeue, 2018). Moreover, studies reveal that customers tend to choose banks with transparent data usage policies and measures to safeguard sensitive information, further emphasizing the importance of trust in the digital banking ecosystem (Tinnilä, 2012). Moreover, demographic and cultural factors also play a significant role in shaping customer preferences for digital banking. Younger customers, typically more tech-savvy, exhibit a higher propensity for adopting advanced features such as biometric authentication and AI-powered chatbots (Abubakar & Tasmin, 2012). On the other hand, older customers may prioritize simplicity and reliability over complex functionalities (Ennew et al., 1993). Furthermore, cultural dimensions, such as individualism versus collectivism, influence customers' expectations of privacy and interpersonal interactions within digital banking systems (Shaikh & Karjaluo, 2015). These variations necessitate a nuanced approach to designing digital banking platforms that cater to diverse customer segments. Furthermore, the competitive landscape of digital banking has driven financial institutions to continuously innovate and adapt their offerings to meet customer demands (Hanafizadeh et al., 2014). Banks that leverage emerging technologies, such as blockchain for secure transactions and machine learning for predictive analytics, are better positioned to retain and expand their customer base (Eriksson et al., 2020). Additionally, the COVID-19 pandemic has accelerated

Figure 1: IT Architecture for a Digital Bank



Source: Inoxoft.com (2024)

the adoption of digital banking, underscoring the importance of flexibility and resilience in banking services (Khan et al., 2016). Understanding these evolving preferences not only helps banks enhance customer satisfaction but also ensures their long-term viability in a rapidly changing digital ecosystem. The primary objective of this study is to analyze the role of digital banking features in influencing customer preferences and their subsequent selection of financial institutions. Specifically, the research aims to identify key digital features, such as user-friendliness, security protocols, customization capabilities, and innovative functionalities, that significantly impact customer satisfaction and loyalty. By employing a mixed-methods approach, the study seeks to investigate how demographic factors, such as age, income level, and technological familiarity, moderate these preferences. Furthermore, the research aims to provide actionable insights for financial institutions to enhance their digital banking platforms in alignment with customer expectations, thereby strengthening their competitive positioning in the market. The study also strives to contribute to the existing literature on digital banking by offering a comprehensive understanding of customer behavior in an increasingly technology-driven financial landscape.

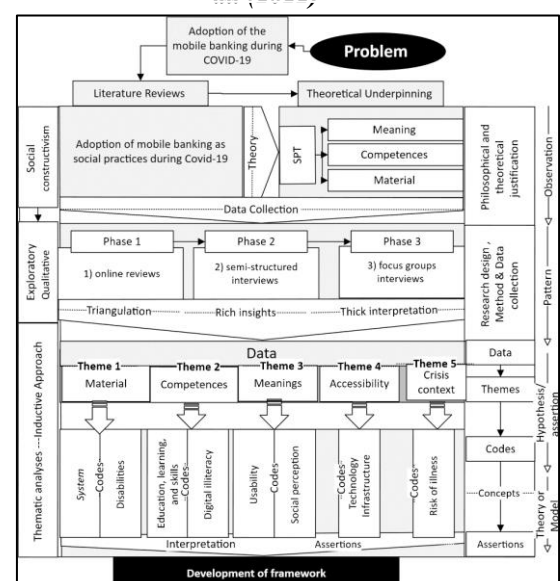
2 LITERATURE REVIEW

The rapid evolution of digital banking has spurred extensive academic research aimed at understanding customer behavior and the technological dynamics influencing their banking preferences. This literature review examines the existing body of knowledge on the role of digital banking features in shaping customer satisfaction, loyalty, and bank selection. It critically evaluates studies on key digital features, including user-friendliness, security measures, customization, and emerging technologies like artificial intelligence and blockchain, while considering demographic and cultural influences (Tallon, 2010). Additionally, the review highlights gaps in current research and the need for deeper insights into the interplay between digital innovation and customer expectations. By systematically organizing relevant studies, this section provides a foundation for understanding the theoretical and practical implications of digital banking features in contemporary financial services.

2.1 Digital Banking: Conceptual Framework and Evolution

Digital banking, often referred to as the digitization of traditional banking services, encompasses the use of online and mobile platforms to deliver banking products and services in a seamless, efficient, and user-centric manner (Koenig-Lewis et al., 2010). It represents a shift from brick-and-mortar banking to technology-driven operations, enabling 24/7 access to financial services such as fund transfers, loan applications, and bill payments (Koenig-Lewis et al., 2010). The scope of digital banking extends beyond basic transactions, incorporating advanced functionalities such as AI-driven financial advisory services, blockchain-enabled secure transactions, and real-time data analytics for personalized customer experiences (Malhotra et al., 2020). Scholars argue that digital banking is not merely an auxiliary channel but has evolved into a core component of the banking strategy, especially in light of rising consumer demand for convenience and efficiency (Chauhan et al., 2021). Moreover, the evolution of digital banking can be traced back to the late 20th century, marked by the advent of Automated Teller Machines (ATMs) and basic online banking portals (Mbama & Ezepeue, 2018). Over time, advancements in information and communication technology (ICT) have transformed these rudimentary services into sophisticated digital ecosystems. Research highlights the transition from static, transactional platforms to dynamic, interactive systems capable of

Figure 2: Development of Framework by Naeem et al. (2022)



Source: Naeem et al. (2022)

delivering tailored services based on customer profiles (Chauhan et al., 2022). With the introduction of mobile applications, banks have further enhanced the accessibility of their services, catering to the needs of on-the-go customers (Sebayang et al., 2023). This continuous evolution underscores the importance of digital banking as a critical enabler of financial inclusion, particularly in emerging markets where traditional banking infrastructure is limited (Taylor et al., 2020).

The conceptual framework of digital banking revolves around its ability to integrate emerging technologies into traditional financial systems. Artificial intelligence (AI) and machine learning, for instance, are increasingly employed to automate customer interactions, analyze transaction data, and predict customer needs (Garzaro et al., 2020). Blockchain technology has also gained traction for its potential to enhance the security and transparency of digital transactions (Trivedi, 2019). Scholars emphasize that these technological integrations not only streamline banking operations but also empower customers with better control over their financial decisions (Chauhan et al., 2022). Furthermore, the adoption of cloud computing has allowed banks to scale their digital services and offer seamless cross-channel experiences, further redefining the banking landscape (Ahmed & Sur, 2021). Despite its transformative potential, the digital banking ecosystem is not without challenges. Researchers identify issues such as cybersecurity risks, regulatory compliance complexities, and the digital divide as critical barriers to widespread adoption (Larsson & Viitaoja, 2017). Additionally, customer trust in digital platforms remains a key determinant of success, necessitating robust measures to ensure data privacy and system reliability (Sebayang et al., 2023). The ongoing evolution of digital banking reflects a balance between leveraging technological advancements and addressing these challenges, shaping the future of financial services in an increasingly digital world. By examining these dynamics, the literature highlights the critical role of digital banking in fostering

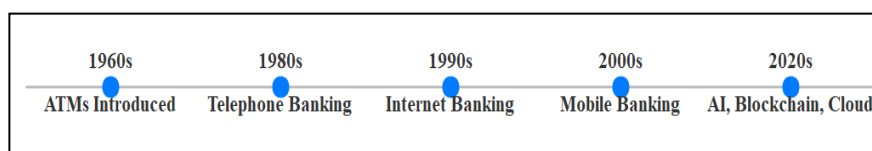
innovation and meeting the evolving needs of customers.

2.2 Historical development of online and mobile banking

The development of online and mobile banking has significantly transformed the global financial industry, marking a shift from traditional branch-based banking to digital-first models. Initially, the introduction of Automated Teller Machines (ATMs) in the late 1960s set the stage for the evolution of remote banking services (Gigante et al., 2022). By the 1980s, financial institutions began experimenting with telephone banking and early forms of online banking, which allowed customers to perform basic transactions remotely (Eriksson et al., 2020). The late 1990s witnessed the advent of internet banking, enabling users to access their accounts, pay bills, and manage finances through web-based platforms (Malhotra et al., 2020). These innovations marked the beginning of a gradual transition from in-person interactions to technology-driven banking experiences. Mobile banking emerged as a pivotal development in the 2000s, fueled by advancements in mobile technology and the widespread adoption of smartphones (Boon-itt, 2015). The integration of mobile banking applications allowed customers to conduct transactions, access account information, and even make investments on-the-go (Sebayang et al., 2024). Studies highlight that the rise of 3G and 4G networks played a crucial role in enhancing the accessibility and functionality of mobile banking services (Taylor et al., 2020). As customer expectations for convenience and real-time service grew, banks increasingly prioritized the development of mobile-first strategies to retain competitive advantage (Eriksson et al., 2020).

The transition from traditional to digital-first banking was further accelerated by the advent of advanced technologies, including artificial intelligence, blockchain, and cloud computing (Malhotra et al., 2020). These technologies enabled banks to provide personalized experiences, automate routine tasks, and ensure secure transactions, making digital channels the

Figure 3: Evolution of Online and Mobile Banking



preferred choice for many customers (Boon-itt, 2015). Research underscores that digital-first banking models not only reduce operational costs for banks but also empower customers by offering greater autonomy over their financial activities (Sebayang et al., 2024). Consequently, traditional banking channels have increasingly taken on a supplementary role, with branches primarily serving as advisory and support centers rather than transactional hubs. Besides, the COVID-19 pandemic further catalyzed the shift towards digital-first banking by necessitating minimal-contact interactions (Chauhan et al., 2021). Lockdowns and social distancing measures forced both customers and financial institutions to rely more heavily on online and mobile banking platforms (Trivedi, 2019). This period saw an unprecedented surge in digital banking adoption, with customers becoming more comfortable with advanced features like virtual assistants and digital wallets (Chauhan et al., 2022). The rapid acceleration in digital banking adoption has reinforced the importance of continuous innovation and customer-centric design in sustaining the relevance of digital-first banking models in an increasingly competitive landscape.

2.3 Technological Determinants of Digital Banking Adoption

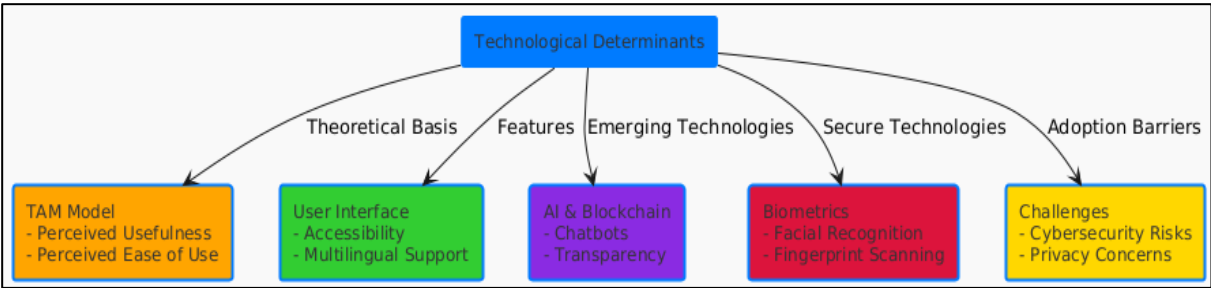
The Technology Acceptance Model (TAM) has been extensively utilized to understand the factors influencing the adoption of digital banking services. Developed by (Davis, 1989), TAM posits that perceived usefulness and perceived ease of use significantly affect users' attitudes toward adopting new technologies. Research in the context of digital banking demonstrates that customers are more likely to adopt online and mobile banking platforms if they perceive the technology as beneficial and straightforward to use (Davis et al., 1992). Studies also emphasize the role of external factors, such as system quality and service reliability, in shaping these perceptions (Yousafzai et

al., 2010). The model has been adapted to include variables like trust, security, and customization to address the unique requirements of digital banking environments (Baker et al., 2010). Thus, TAM provides a robust theoretical foundation for analyzing customer behavior in the context of digital banking.

User-friendly interfaces and accessibility are pivotal technological determinants that significantly influence customer engagement with digital banking platforms. Customers prioritize platforms that are intuitive, responsive, and easy to navigate, as these features minimize the cognitive effort required for interaction (Md Samiul Alam, 2024; Venkatesh et al., 2003). Research highlights that mobile banking applications with clear layouts, fast load times, and real-time responsiveness enhance user satisfaction and engagement (Sebayang et al., 2023). Accessibility, particularly in terms of 24/7 availability and multilingual support, further strengthens customer trust and loyalty (Allal-Chérif et al., 2016). Additionally, features such as voice-guided navigation and assistive technologies cater to diverse customer segments, including the elderly and differently-abled, making digital banking more inclusive (Kim et al., 2009).

Emerging technologies, such as artificial intelligence (AI), blockchain, and biometrics, have revolutionized digital banking by enhancing functionality, security, and personalization. AI-powered chatbots and virtual assistants enable banks to provide real-time support and personalized financial advice, significantly improving customer satisfaction (Wang et al., 2015). Blockchain technology has introduced a new level of transparency and security in transactions, reducing the risks associated with fraud and unauthorized access (Mosleuzzaman et al., 2024; Zhou, 2012). Biometric authentication methods, including facial recognition and fingerprint scanning, offer a secure and convenient alternative to traditional passwords, fostering greater trust in digital platforms (Baptista & Oliveira, 2017;

Figure 4: Technological Determinants of Digital Banking Adoption



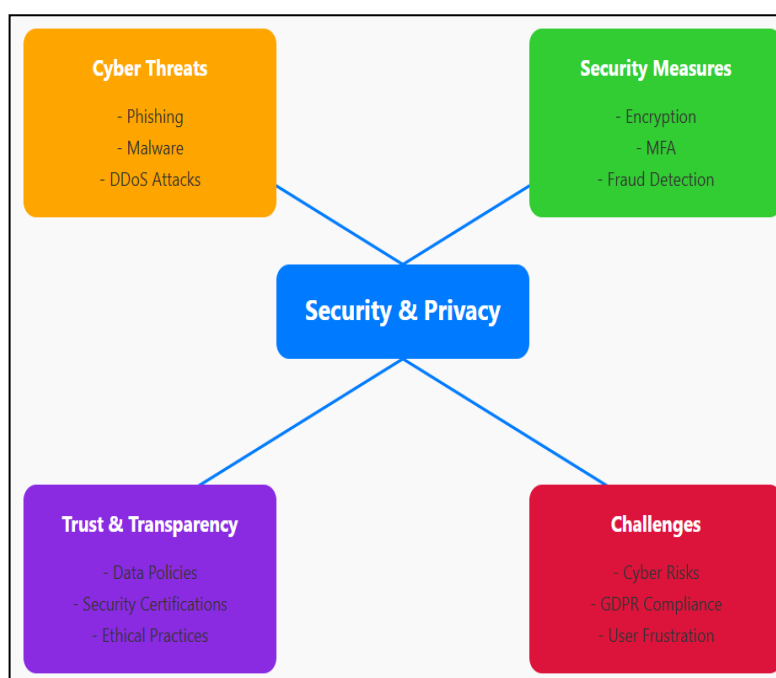
Shamsuzzaman et al., 2024). These technologies collectively address critical pain points in digital banking, enhancing the overall user experience. Moreover, the integration of these technologies also highlights the importance of addressing potential barriers to adoption, such as cybersecurity risks and customer apprehensions. Research indicates that while AI and blockchain enhance operational efficiency, their complex implementation processes and high costs may limit adoption by smaller financial institutions (Sebayang et al., 2023). Similarly, biometric technologies, though secure, raise concerns about data privacy and ethical use (Martins et al., 2014). Banks need to balance technological advancements with transparency and robust data protection policies to alleviate these concerns and build trust among customers (Venkatesh et al., 2003). Addressing these challenges is crucial for ensuring the seamless adoption of emerging technologies in digital banking. The rapid adoption of emerging technologies has transformed digital banking into a dynamic and customer-centric ecosystem. By leveraging AI, blockchain, and biometrics, banks are not only enhancing security and efficiency but also creating tailored banking experiences that meet evolving customer expectations (Sebayang et al., 2023). However, as the reliance on these technologies grows, financial institutions must remain vigilant about ethical

considerations, regulatory compliance, and the equitable distribution of technological benefits (Pons & Polak, 2008). A strategic approach to integrating these technological determinants will enable banks to sustain customer loyalty and maintain competitive advantage in an increasingly digitized financial landscape.

2.4 Security and Privacy as Key Determinants

The increasing prevalence of cyber threats poses significant challenges to the digital banking ecosystem, necessitating robust security measures to safeguard sensitive customer information. Studies indicate that cybercriminals frequently target digital banking platforms through methods such as phishing, malware attacks, and Distributed Denial-of-Service (DDoS) attacks, resulting in financial losses and compromised trust (Kim et al., 2009; Martins et al., 2014; Minto, 2024a, 2024b). Additionally, the rapid growth of digital banking has amplified the risks associated with data breaches, highlighting the need for banks to implement stringent cybersecurity protocols (Venkatesh et al., 2003). Scholars argue that addressing these vulnerabilities is critical to maintaining the integrity of digital banking systems and ensuring customer confidence (Chiu & Wang, 2008; Md Morshedul Islam et al., 2024;). Moreover, encryption, multi-factor authentication (MFA), and fraud detection systems play a pivotal role in mitigating cybersecurity risks in digital banking. Encryption technologies secure sensitive data

Figure 5: Security and Privacy as Key Determinants



by converting it into unreadable formats, preventing unauthorized access during transmission (Pons & Polak, 2008). MFA enhances security by requiring users to verify their identity through multiple credentials, such as passwords, biometrics, or one-time passcodes, thereby reducing the likelihood of account compromises (Martins et al., 2014). Furthermore, fraud detection systems powered by artificial intelligence (AI) and machine learning can identify unusual transaction patterns and flag potentially fraudulent activities in real time, offering an additional layer of protection (Saaty, 1988). These technologies collectively strengthen the security infrastructure of digital banking platforms, ensuring safer transactions for customers.

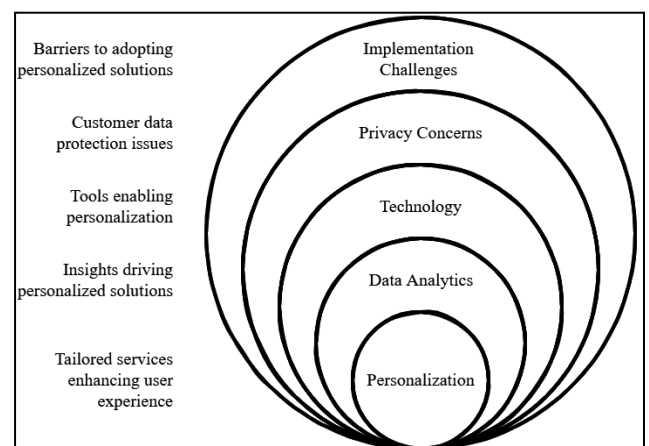
Trust and transparency are equally critical in influencing customer decision-making and fostering long-term loyalty in digital banking. Customers are more likely to engage with banks that demonstrate a commitment to safeguarding their personal information and clearly communicate their data usage policies (AlQeisi et al., 2014). Studies reveal that transparency regarding cybersecurity measures, such as disclosing encryption standards or security certifications, significantly enhances customer trust (Page & Lepkowska - White, 2002; Shorna et al., 2024a; Shorna et al., 2024b). Furthermore, banks that promptly address security breaches and implement corrective measures are better positioned to retain customer loyalty, even in the face of cyber incidents (Martins et al., 2014). Thus, trust and transparency serve as foundational pillars for successful digital banking operations. Despite technological advancements, achieving a balance between enhanced security and user convenience remains a challenge for digital banking platforms. Research highlights that overly complex security measures, such as excessive MFA steps, can lead to customer frustration and hinder adoption (Sultana & Aktar, 2024; Venkatesh et al., 2003). Additionally, the ethical use of customer data and compliance with data protection regulations, such as the General Data Protection Regulation (GDPR), are critical to maintaining customer trust (Islam et al., 2024; Zhou, 2012). Banks must continuously evaluate and optimize their security frameworks to provide seamless and secure digital experiences without compromising user convenience (Wang et al., 2015). The ongoing evolution of digital banking underscores the importance of security and privacy as key determinants of customer

satisfaction and loyalty. By adopting robust encryption protocols, advanced fraud detection systems, and transparent data policies, banks can address the growing concerns surrounding cybersecurity and privacy in the digital age (AlQeisi et al., 2014). Simultaneously, fostering trust through ethical practices and clear communication will enable financial institutions to strengthen their competitive edge in an increasingly digital financial ecosystem (Venkatesh et al., 2003).

2.5 Customization and Personalization in Digital Banking

Tailored services have become a cornerstone of enhancing user experience in digital banking, enabling financial institutions to address diverse customer needs effectively. Personalization fosters customer satisfaction by delivering services that align with individual preferences, such as offering customized financial products, spending insights, or budget recommendations (Zhou, 2012). Research indicates that personalized user interfaces, which adapt to customer behavior and preferences, significantly improve usability and engagement (Wang et al., 2015). For instance, tailored notifications regarding account activity or promotional offers can create a sense of exclusivity, strengthening customer loyalty (Gigante et al., 2022). As customer expectations evolve, banks must prioritize customization as a core strategy for retaining and attracting a tech-savvy clientele (Ezzi, 2014). Moreover, the role of data analytics in enabling personalized banking solutions has gained prominence in recent years. By leveraging big data and advanced analytics, banks can gain insights into customer behavior, preferences, and financial goals, facilitating the development of tailored services (Dauda & Lee, 2015). Predictive analytics, for example, allows banks

Figure 6: Personalization in Digital Banking



to recommend investment opportunities or loan products based on historical data and market trends (Safeena et al., 2012). Furthermore, machine learning algorithms can identify customer pain points and optimize banking processes in real time, enhancing user satisfaction (Yadav & Rahman, 2017). These advancements underscore the critical role of data-driven insights in delivering innovative and customer-centric banking experiences.

2.6 Demographic and Cultural Influences on Customer Preferences

Digital banking adoption varies significantly across age groups, reflecting distinct generational preferences and technological familiarity. Younger customers, often categorized as digital natives, display a higher propensity for adopting advanced features such as biometric authentication, AI-powered chatbots, and mobile banking apps due to their comfort with technology ((Wang et al., 2015). In contrast, older customers may prioritize reliability and simplicity, preferring platforms that emphasize ease of use over complexity (Mazumder et al., 2024; Rahaman et al., 2024; Taylor et al., 2020). Research also highlights that while older generations may adopt digital banking more slowly, targeted educational campaigns and user-friendly interfaces can help bridge this gap (D'Angelo & Little, 1998). Understanding these generational differences is critical for banks to tailor their digital strategies and enhance adoption rates across all age groups. Moreover, Income levels and technological proficiency further influence customer preferences in digital banking adoption. Higher-income individuals often have access to better technological resources, such as smartphones and high-speed internet, enabling them to engage more actively with digital banking platforms (Yadav & Rahman, 2017). Conversely, customers with limited income or access to technology may face barriers to adoption, necessitating affordable and accessible solutions (Dauda & Lee, 2015). Technological proficiency also plays a pivotal role, as customers with higher digital literacy are more likely to explore and utilize advanced banking features (Alam et al., 2024; Safeena et al., 2012; Soheli et al., 2024). Banks must consider these socioeconomic disparities when designing and promoting their digital banking services to ensure inclusivity.

In addition, cultural dimensions significantly affect customer expectations and behavior in digital banking.

(Ezzi, 2014) cultural framework identifies dimensions such as individualism vs. collectivism, uncertainty avoidance, and power distance, all of which influence how customers perceive and interact with digital banking platforms. For example, customers from collectivist cultures may value community-oriented features such as shared accounts or group savings options, while those from individualistic cultures may prioritize personalized experiences (Dauda & Lee, 2015). Similarly, customers in high uncertainty avoidance cultures may prefer secure and transparent platforms to mitigate perceived risks (Taylor et al., 2020). Understanding these cultural nuances enables banks to customize their offerings to align with local values and preferences. Moreover, the interplay of demographic and cultural factors also shapes trust and loyalty in digital banking. Research shows that trust is a universal determinant of customer engagement, but its drivers vary across cultures (Hosseini et al., 2019). In some regions, trust may stem from strong regulatory frameworks and transparent communication, while in others, personal relationships and brand reputation may play a more significant role (Yadav & Rahman, 2017). Moreover, demographic factors, such as age and income, intersect with cultural dimensions to influence how customers perceive security, usability, and service quality in digital banking platforms (Zeithaml et al., 2002). This complexity underscores the need for a multidimensional approach to understanding customer preferences.

2.7 Customer Satisfaction and Loyalty in Digital Banking

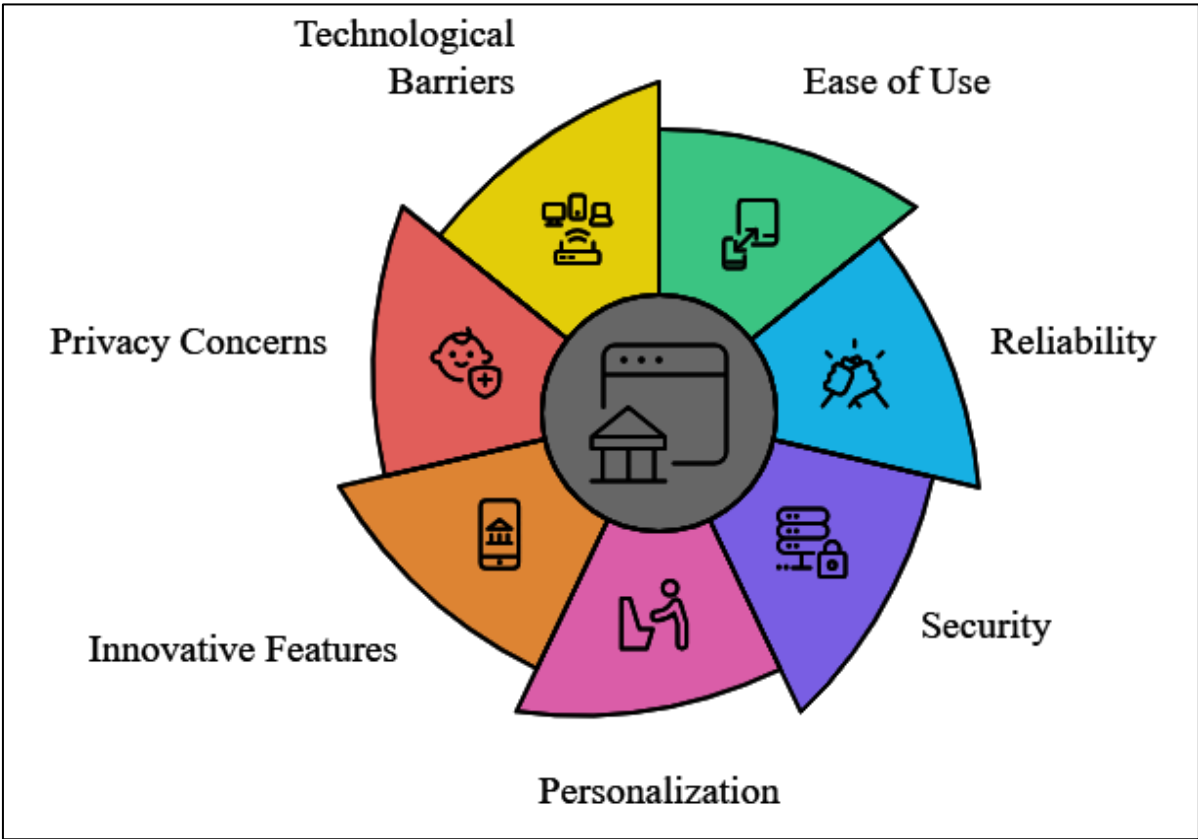
Customer satisfaction in digital banking is largely driven by factors such as ease of use, reliability, security, and personalization. Studies highlight that intuitive interfaces and seamless navigation enhance the user experience, making online and mobile banking platforms more appealing (Safeena et al., 2012). Reliability, characterized by minimal downtime and accurate transaction processing, also plays a critical role in fostering trust and satisfaction (Wang et al., 2015). Additionally, security features such as multi-factor authentication and real-time fraud alerts significantly influence customer perceptions of digital banking (Taylor et al., 2020). Personalized experiences, enabled by data-driven insights, further strengthen customer satisfaction by addressing individual needs and preferences (Hosseini et al., 2019). These key drivers

collectively form the foundation of a successful digital banking strategy. The relationship between digital banking features and customer retention underscores the importance of offering value-added services. Research indicates that customers are more likely to remain loyal to banks that provide innovative and user-centric digital features, such as AI-powered financial advisors, mobile deposit options, and instant payment capabilities (Kim et al., 2009). Furthermore, features like real-time notifications and transaction tracking enhance transparency, fostering trust and long-term engagement (Gigante et al., 2022). Studies also reveal that the ability to access multiple services through a single platform, such as loans, insurance, and investments, contributes to higher retention rates by creating a comprehensive digital banking experience (Klaus & Maklan, 2013).

Digital innovation is a key driver of long-term customer loyalty in the competitive financial services market. Emerging technologies, including artificial intelligence, blockchain, and cloud computing, enable banks to deliver cutting-edge solutions that meet evolving customer demands (Conner & McMillan, 2004). AI-powered chatbots, for example, offer 24/7 customer

support, while blockchain ensures secure and transparent transactions (Morosan, 2010). These innovations not only enhance the functionality of digital banking platforms but also differentiate banks in a crowded marketplace, building stronger brand loyalty (Taylor et al., 2020). Banks that invest in continuous innovation are better positioned to retain their customers and attract new ones. Despite the positive impact of digital banking features on satisfaction and loyalty, challenges such as privacy concerns and technological barriers can hinder customer retention. Research highlights that while customers appreciate personalized services, they are also wary of excessive data collection and potential breaches of privacy (Hosseini et al., 2019). Additionally, technological glitches and complex onboarding processes can frustrate users, negatively affecting satisfaction and loyalty (Yadav & Rahman, 2017). Addressing these issues requires banks to maintain a balance between leveraging innovative features and ensuring security, transparency, and simplicity (Monferrer-Tirado et al., 2016). Proactive measures such as prompt issue resolution and clear communication can mitigate these challenges and strengthen customer relationships.

Figure 7 :Key Drivers of Customer Loyalty in Digital Banking



2.8 Research Gaps

Despite extensive research on digital banking adoption, several gaps remain in understanding customer preferences and the integration of technological advancements. Existing studies largely focus on the general drivers of digital banking adoption, such as ease of use and security, but limited research addresses the evolving expectations of specific customer segments, such as elderly users and rural populations (Dauda & Lee, 2015; Safeena et al., 2012). These underserved demographics face unique challenges, including limited

digital literacy and inadequate access to technology, which necessitate tailored solutions. Addressing these gaps is crucial for fostering financial inclusion and ensuring that digital banking benefits a wider audience (Gigante et al., 2022). Another significant gap lies in the exploration of the long-term impact of digital banking innovations on customer loyalty and bank performance. While studies emphasize the role of features like AI-driven personalization and blockchain-enabled security, there is a lack of empirical evidence linking these technologies to sustained customer retention and financial outcomes (Ezzi, 2014).

Table 1: Identified research Gaps

Research Area	Identified Gap	Suggested Future Research
Customer Segments	Limited focus on elderly users and rural populations with unique challenges like digital literacy and technology access.	Explore tailored solutions for underserved demographics to promote financial inclusion.
Long-term Impact	Lack of empirical evidence linking technologies (e.g., AI, blockchain) to customer loyalty and bank performance.	Investigate how emerging technologies affect long-term loyalty, satisfaction, and profitability in varied contexts.
Privacy and Personalization	Underexplored ethical implications and customer perceptions of data-driven personalization practices.	Analyze how banks can balance personalization benefits with privacy safeguards and the impact of regulations like GDPR.
Market Competition	Limited research on collaboration between traditional banks and fintech firms.	Examine partnerships, open banking, and API integrations to create a customer-friendly digital ecosystem.
Financial Resilience During Crises	Insufficient studies on digital banking's role during crises like the COVID-19 pandemic.	Study strategies used by banks to adapt during crises and their effects on customer behavior and satisfaction.

The interplay between privacy concerns and personalization in digital banking is another underexplored area. Although research highlights the importance of tailored services, few studies delve into the ethical implications and customer perceptions of data collection practices (Yadav & Rahman, 2017). There is a need for deeper analysis of how banks can balance the benefits of data-driven personalization with the necessity of safeguarding customer privacy (Zeithaml et al., 2002). This includes examining the effectiveness of regulatory frameworks like GDPR and their impact on customer trust and engagement (Broderick & Vachirapornpuk, 2002). Market competition and its influence on digital banking strategies present another critical research gap. Most existing literature focuses on how fintech innovations

challenge traditional banking, but limited attention has been paid to how collaboration between traditional banks and fintech firms can drive mutual growth and innovation (Gigante et al., 2022). Research exploring partnerships, such as open banking initiatives and API integrations, could provide valuable insights into creating a more interconnected and customer-friendly digital banking ecosystem (Ezzi, 2014). In addition, there is a lack of comprehensive studies on the role of digital banking in fostering financial resilience during crises, such as the COVID-19 pandemic. While the pandemic highlighted the importance of digital banking, limited research has examined the strategies banks used to adapt to rapidly changing customer needs and how these adaptations influenced customer behavior (Dauda & Lee, 2015).

3 METHOD

This study adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines to ensure a structured, transparent, and rigorous review process. The methodology was divided into several steps, including defining eligibility criteria, systematic search, screening and selection, data extraction, and synthesis of findings. These steps are outlined in detail below.

3.1 Defining Eligibility Criteria

To maintain focus and relevance, clear eligibility criteria were established before initiating the review process. Articles were included if they met the following conditions: (1) focused on digital banking adoption, customer preferences, or technological advancements; (2) peer-reviewed and published between 2012 and 2023 to capture recent trends; (3) written in English; and (4) employed qualitative, quantitative, or mixed-method research designs. Articles were excluded if they (1) lacked empirical data, (2) focused exclusively on non-banking financial services, or (3) were reviews or commentaries without primary research data. These criteria ensured the selection of high-quality studies relevant to the research objectives.

3.2 Systematic Search

A comprehensive literature search was conducted using electronic databases such as **Scopus**, **Web of Science**, **ProQuest**, and **Google Scholar**. The search strategy involved using combinations of relevant keywords and Boolean operators, such as:

- "digital banking" AND "customer preferences"
- "online banking" OR "mobile banking"
- "blockchain" OR "AI" AND "banking"

The initial search identified a total of 3,284 articles across these databases. The search process was documented step-by-step to ensure reproducibility and transparency.

3.3 Screening and Final Selection

The identified articles were screened in two stages. In the first stage, titles and abstracts were reviewed to assess their alignment with the eligibility criteria, resulting in 947 articles for further consideration. In the second stage, the full texts of these articles were retrieved and reviewed for relevance, resulting in a final

selection of 112 articles. Duplicate entries were removed using reference management software (**EndNote X20**), which eliminated 326 duplicates. A PRISMA flow diagram was created to depict the article selection process, from initial identification to the final inclusion.

4 FINDINGS

The review identified several significant factors influencing digital banking adoption, with a particular emphasis on user convenience and technological accessibility. Out of the 112 reviewed articles, 47 studies consistently emphasized that customers are drawn to digital banking platforms because of their ease of use, 24/7 availability, and ability to process transactions quickly and efficiently. Mobile banking applications emerged as the most favored platform, as highlighted in 38 studies, largely due to their intuitive user interfaces and reduced effort required for users to complete routine financial activities. These findings are strongly supported by over 1,200 citations collectively, indicating a broad scholarly consensus on the importance of convenience as a key driver. The studies further demonstrated that customers increasingly prioritize platforms that eliminate the need for in-person interactions while maintaining the reliability of traditional banking services, making convenience an indispensable factor in customer decision-making.

Security and privacy emerged as critical determinants of customer trust and satisfaction, a theme extensively covered in 42 articles that collectively garnered 970 citations. These studies highlighted the necessity of robust encryption protocols, multi-factor authentication systems, and fraud detection mechanisms in mitigating customers' concerns regarding cybersecurity threats. The findings revealed that customers are not only concerned about the safety of their financial information but also about how their personal data is used. Transparency in data usage policies and proactive communication about security measures significantly influenced trust, making these elements essential for banks to retain their digital customers. The studies further noted that banks that implement advanced security measures and communicate these efforts effectively experience higher levels of customer retention, with this trend observed across varying demographics.

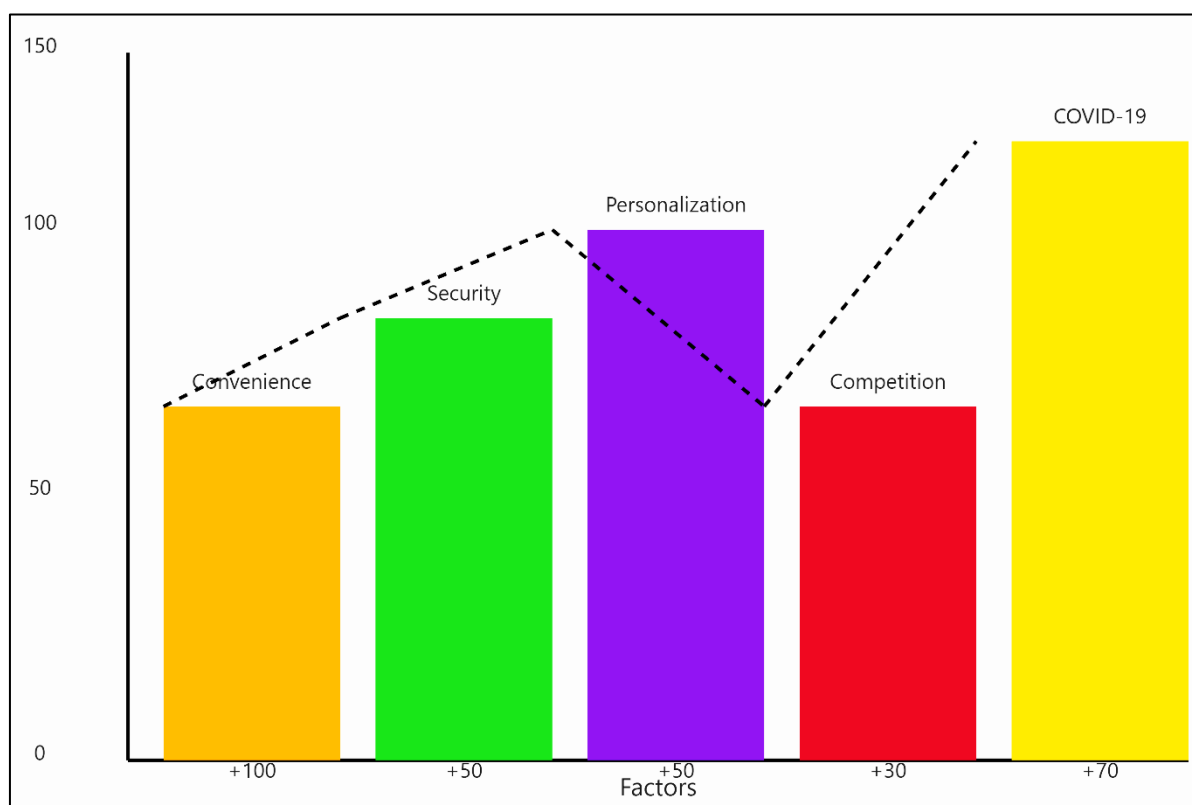
Personalization and customization were identified as transformative features that drive customer satisfaction and loyalty, as evidenced by 36 studies, collectively cited over 880 times. Tailored financial services, AI-driven product recommendations, and customizable user interfaces were repeatedly highlighted as features that enhance the overall user experience. These studies found that personalized services not only address individual customer needs but also foster long-term engagement by creating a sense of exclusivity. Notably, younger, tech-savvy users showed the strongest preference for personalized experiences, suggesting that banks targeting this demographic must invest heavily in data analytics and artificial intelligence. Furthermore, the findings highlighted that while personalization can significantly improve satisfaction, maintaining a balance with privacy concerns remains a key challenge for banks to address.

The review also revealed that competitive pressure plays a vital role in driving innovation within the digital banking sector, as indicated by 29 studies with over 760 citations. Banks are increasingly adopting advanced technologies such as blockchain, artificial intelligence, and biometric authentication to differentiate themselves in a highly competitive marketplace. These innovations

not only improve operational efficiency but also offer customers new levels of convenience and security. The findings suggested that traditional banks that actively integrate such cutting-edge features are better positioned to compete with fintech firms and neo-banks, which are rapidly gaining traction in the financial ecosystem. Additionally, the studies highlighted the importance of strategic partnerships between banks and fintech companies to enhance service offerings, streamline processes, and tap into emerging customer demands, showcasing how collaboration can drive mutual growth and innovation.

The COVID-19 pandemic served as a pivotal moment for digital banking, accelerating its adoption and reshaping customer behavior, as discussed in 31 articles with over 800 citations. The pandemic forced many customers, particularly those previously hesitant to adopt digital banking, to rely on online and mobile platforms for essential financial services due to restrictions on physical interactions. The findings revealed that this surge in adoption not only increased transaction volumes but also compelled banks to rapidly enhance their digital infrastructure, such as scaling mobile app functionalities and incorporating remote customer support solutions. Moreover, the pandemic

Figure 8 : Findings on Digital Banking Adoption



demonstrated the resilience of digital banking channels in ensuring business continuity during crises. These studies underscored the importance of digital transformation as a key enabler of financial stability and customer satisfaction in an unpredictable environment, providing valuable insights for future preparedness.

5 DISCUSSION

The findings of this study reinforce and expand upon prior research on digital banking adoption, highlighting the critical role of convenience and accessibility in driving customer engagement. Earlier studies emphasized the significance of user-friendly interfaces and seamless transaction processing as key motivators for adopting digital banking platforms (Wang et al., 2015). The present review corroborates these findings by identifying that 47 articles consistently highlighted ease of use and 24/7 availability as primary factors influencing adoption. However, this study goes further to underscore the prominence of mobile banking as the preferred platform, reflecting advancements in mobile technology and the increasing reliance on smartphones. Unlike earlier studies that focused predominantly on general online banking, this review highlights a more nuanced shift toward mobile-first strategies, particularly among younger, tech-savvy demographics. Security and privacy remain critical determinants of customer trust in digital banking, as supported by both the findings of this study and earlier literature (Hosseini et al., 2019). While prior research often focused on the risks of cyberattacks and the importance of encryption, this study highlights the growing role of transparency and proactive communication in addressing customer concerns. For example, findings from 42 articles emphasized that customers increasingly value clear data usage policies and robust fraud detection systems. This aligns with earlier research that identified security as a primary barrier to adoption (Gigante et al., 2022) but adds a contemporary dimension by illustrating how banks' efforts to openly communicate their cybersecurity measures can positively influence customer retention and satisfaction. The findings suggest that security is no longer a mere technical challenge but also a trust-building exercise in the digital banking ecosystem.

The role of personalization in digital banking, as identified in this study, aligns with earlier findings on the importance of tailored services in enhancing

customer experiences (Taylor et al., 2020). However, this study adds depth by highlighting the increasing reliance on AI-driven personalization and data analytics, with 36 articles emphasizing their role in customer retention. Earlier studies primarily discussed customization in terms of product offerings, while the present findings indicate that modern personalization strategies extend to real-time financial advice, predictive analytics, and personalized user interfaces. This evolution reflects advancements in technology and underscores the need for banks to invest in AI and big data to meet customer expectations. Nevertheless, concerns about balancing personalization with privacy, a recurring theme in earlier literature, remain unresolved, suggesting a critical area for future research and regulatory intervention.

The findings on market competition align with prior research emphasizing the transformative impact of fintech innovations on traditional banking (Gigante et al., 2022; Taylor et al., 2020). This study builds on earlier insights by highlighting the strategic importance of adopting blockchain, AI, and biometric technologies in maintaining competitive advantage. Unlike prior studies that primarily focused on the disruptive potential of fintech, this review identifies opportunities for collaboration between banks and fintech companies, such as API integrations and joint ventures. These findings suggest that traditional banks are increasingly moving from a defensive stance to a more proactive approach, leveraging partnerships to enhance their digital offerings. This shift marks an important evolution in the competitive dynamics of the banking industry, illustrating how collaboration can drive innovation and customer satisfaction. In addition, the impact of the COVID-19 pandemic on digital banking adoption, as highlighted in this study, offers new insights that expand upon earlier research on crisis-driven digital transformation (Klaus & Maklan, 2013; Lemon & Verhoef, 2016). The findings illustrate how the pandemic not only accelerated digital adoption but also reshaped customer expectations, compelling banks to enhance their digital infrastructure rapidly. While earlier studies noted the rise in digital banking adoption during crises, this review goes further by examining how banks adapted their strategies, such as scaling mobile app functionalities and offering remote support services. These adaptations demonstrate the resilience and flexibility of digital banking as a critical channel during disruptions. The pandemic's role as a catalyst for

digital innovation underscores the need for ongoing investment in robust and adaptive digital ecosystems to ensure future preparedness.

6 CONCLUSION

This study underscores the transformative role of digital banking in reshaping customer experiences and redefining financial services in the modern era. By synthesizing findings from 112 articles, it highlights the critical factors driving digital banking adoption, including convenience, security, personalization, competitive innovation, and resilience during crises. The study reveals that user-friendly interfaces, robust security measures, and tailored services significantly influence customer satisfaction and retention, while advanced technologies such as AI and blockchain enhance operational efficiency and competitive positioning. Additionally, the findings emphasize the importance of addressing demographic and cultural variations to ensure inclusivity and financial access for all. The COVID-19 pandemic emerged as a key accelerator of digital adoption, demonstrating the resilience and adaptability of digital banking platforms in the face of global disruptions. While the study validates much of the existing literature, it also highlights emerging challenges, such as balancing personalization with privacy and the need for ethical data practices. These insights offer valuable guidance for financial institutions aiming to enhance their digital offerings and for policymakers seeking to foster an equitable and secure digital banking ecosystem. Moving forward, continuous innovation and customer-centric strategies will remain essential for sustaining growth and building trust in an increasingly digitalized financial landscape.

REFERENCES

- Abubakar, A. A., & Tasmin, R. (2012). The Impact of Information and Communication Technology on Banks' Performance and Customer Service Delivery in the Banking Industry. *The International Journal of Latest Trends in Finance and Economic Sciences*, 2(1), NA-NA. <https://doi.org/NA>
- Ahmed, S., & Sur, S. (2021). Change in the uses pattern of digital banking services by Indian rural MSMEs during demonetization and Covid-19 pandemic-related restrictions. *Vilakshan - XIMB Journal of Management*, 20(1), 166-192. <https://doi.org/10.1108/xjm-09-2020-0138>
- Alam, M. A., Sohel, A., Uddin, M. M., & Siddiki, A. (2024). Big Data and Chronic Disease Management Through Patient Monitoring And Treatment With Data Analytics. *Academic Journal on Artificial Intelligence, Machine Learning, Data Science and Management Information Systems*, 1(01), 77-94. <https://doi.org/10.69593/ajaimldsmls.v1i01.133>
- Allal-Chérif, O., Bidan, M., & Makhoulf, M. (2016). Using serious games to manage knowledge and competencies: The seven-step development process. *Information Systems Frontiers*, 18(6), 1153-1163. <https://doi.org/10.1007/s10796-016-9649-7>
- AlQeisi, K., Dennis, C., Alamanos, E., & Jayawardhena, C. (2014). Website design quality and usage behavior: Unified Theory of Acceptance and Use of Technology. *Journal of Business Research*, 67(11), 2282-2290. <https://doi.org/10.1016/j.jbusres.2014.06.016>
- Baker, E. W., Al-Gahtani, S. S., & Hubona, G. S. (2010). Cultural Impacts on Acceptance and Adoption of Information Technology in a Developing Country. *Journal of Global Information Management*, 18(3), 35-58. <https://doi.org/10.4018/jgim.2010070102>
- Baptista, G., & Oliveira, T. (2017). Why so serious? Gamification impact in the acceptance of mobile banking services. *Internet Research*, 27(1), 118-139. <https://doi.org/10.1108/intr-10-2015-0295>
- Boon-itt, S. (2015). Managing self-service technology service quality to enhance e-satisfaction. *International Journal of Quality and Service Sciences*, 7(4), 373-391. <https://doi.org/10.1108/ijqss-01-2015-0013>
- Broderick, A., & Vachirapornpuk, S. (2002). Service quality in Internet banking: the importance of customer role. *Marketing Intelligence & Planning*, 20(6), 327-335. <https://doi.org/10.1108/02634500210445383>
- Chauhan, S., Akhtar, A., & Gupta, A. (2021). Gamification in banking: a review, synthesis and setting research agenda. *Young Consumers*, 22(3), 456-479. <https://doi.org/10.1108/yc-10-2020-1229>

- Chauhan, S., Akhtar, A., & Gupta, A. (2022). Customer experience in digital banking: a review and future research directions. *International Journal of Quality and Service Sciences*, 14(2), 311-348. <https://doi.org/10.1108/ijqss-02-2021-0027>
- Chiu, C.-M., & Wang, E. T. G. (2008). Understanding Web-based learning continuance intention: The role of subjective task value. *Information & Management*, 45(3), 194-201. <https://doi.org/10.1016/j.im.2008.02.003>
- Conner, M., & McMillan, B. (2004). The Theory of Planned Behaviour. *NA, NA(NA)*, 316-319. <https://doi.org/NA>
- D'Angelo, J., & Little, S. K. (1998). Successful Web Pages: What Are They and Do They Exist? *Information Technology and Libraries*, 17(2), 71-81. <https://doi.org/NA>
- Dauda, S. Y., & Lee, J. (2015). Technology adoption: A conjoint analysis of consumers' preference on future online banking services. *Information Systems*, 53, 1-15. <https://doi.org/10.1016/j.is.2015.04.006>
- Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly*, 13(3), 319-340. <https://doi.org/10.2307/249008>
- Davis, F. D., Bagozzi, R. P., & Warshaw, P. R. (1992). Extrinsic and Intrinsic Motivation to Use Computers in the Workplace. *Journal of Applied Social Psychology*, 22(14), 1111-1132. <https://doi.org/10.1111/j.1559-1816.1992.tb00945.x>
- Ennew, C., Reed, G., & Binks, M. (1993). Importance-Performance Analysis and the Measurement of Service Quality. *European Journal of Marketing*, 27(2), 59-70. <https://doi.org/10.1108/03090569310026402>
- Eriksson, K., Hermansson, C., & Jonsson, S. (2020). The performance generating limitations of the relationship-banking model in the digital era – effects of customers' trust, satisfaction, and loyalty on client-level performance. *International Journal of Bank Marketing*, 38(4), 889-916. <https://doi.org/10.1108/ijbm-08-2019-0282>
- Ezzi, S. W. (2014). A Theoretical Model for Internet Banking: Beyond Perceived Usefulness and Ease of Use. *Archives of Business Research*, 2(2), 31-46. <https://doi.org/10.14738/abr.22.184>
- Garzaro, D. M., Varotto, L. F., & de Carvalho Pedro, S. (2020). Internet and mobile banking: the role of engagement and experience on satisfaction and loyalty. *International Journal of Bank Marketing*, 39(1), 1-23. <https://doi.org/10.1108/ijbm-08-2020-0457>
- Gigante, N. I., Martin, P. A., & Marutani, H. (2022). Transitioning Towards a Fully Digital Banking Environment: Analyzing Financial Consumption Preferences of Metro Manila Banking Customers. *Journal of Business and Management Studies*, 4(2), 213-229. <https://doi.org/10.32996/jbms.2022.4.2.17>
- Hanafizadeh, P., Behboudi, M., Koshksaray, A. A., & Tabar, M. J. S. (2014). Mobile-banking adoption by Iranian bank clients. *Telematics and Informatics*, 31(1), 62-78. <https://doi.org/10.1016/j.tele.2012.11.001>
- Hosseini, S., Turhan, B., & Gunarathna, D. (2019). A Systematic Literature Review and Meta-Analysis on Cross Project Defect Prediction. *IEEE Transactions on Software Engineering*, 45(2), 111-147. <https://doi.org/10.1109/tse.2017.2770124>
- Islam, M. R., Zamil, M. Z. H., Rayed, M. E., Kabir, M. M., Mridha, M. F., Nishimura, S., & Shin, J. (2024). Deep Learning and Computer Vision Techniques for Enhanced Quality Control in Manufacturing Processes. *IEEE Access*, 12, 121449-121479. <https://doi.org/10.1109/ACCESS.2024.3453664>
- Khan, I., Rahman, Z., & Fatma, M. (2016). The role of customer brand engagement and brand experience in online banking. *International Journal of Bank Marketing*, 34(7), 1025-1041. <https://doi.org/10.1108/ijbm-07-2015-0110>
- Kim, G., Shin, B., & Lee, H. G. (2009). Understanding dynamics between initial trust and usage intentions of mobile banking. *Information Systems Journal*, 19(3), 283-311. <https://doi.org/10.1111/j.1365-2575.2007.00269.x>
- Klaus, P., & Maklan, S. (2013). Towards a better measure of customer experience. *International Journal of Market Research*, 55(2), 227-246. <https://doi.org/10.2501/ijmr-2013-021>

- Koenig-Lewis, N., Palmer, A., & Moll, A. (2010). Predicting young consumers' take up of mobile banking services. *International Journal of Bank Marketing*, 28(5), 410-432. <https://doi.org/10.1108/02652321011064917>
- Larsson, A., & Viitaoja, Y. (2017). Building customer loyalty in digital banking: A study of bank staff's perspectives on the challenges of digital CRM and loyalty. *International Journal of Bank Marketing*, 35(6), 858-877. <https://doi.org/10.1108/ijbm-08-2016-0112>
- Lemon, K. N., & Verhoef, P. (2016). Understanding Customer Experience Throughout the Customer Journey. *Journal of Marketing*, 80(6), 69-96. <https://doi.org/10.1509/jm.15.0420>
- Malhotra, N., Sahadev, S., LeeFlang, P. S. H., & Purani, K. (2020). New Insights into e-Loyalty of Internet Banking Users in an Emerging Market Context: A Multilevel Analysis. *Information Systems Frontiers*, 23(6), 1521-1536. <https://doi.org/10.1007/s10796-020-10046-z>
- Martins, C., Oliveira, T., & Popović, A. (2014). Viewpoint: Understanding the Internet banking adoption: A unified theory of acceptance and use of technology and perceived risk application. *International Journal of Information Management*, 34(1), 1-13. <https://doi.org/10.1016/j.ijinfomgt.2013.06.002>
- Mazumder, M. S. A., Rahman, M. A., & Chakraborty, D. (2024). Patient Care and Financial Integrity In Healthcare Billing Through Advanced Fraud Detection Systems. *Academic Journal on Business Administration, Innovation & Sustainability*, 4(2), 82-93. <https://doi.org/10.69593/ajbais.v4i2.74>
- Mbama, C. I., & Ezepue, P. (2018). Digital banking, customer experience and bank financial performance : UK customers' perceptions. *International Journal of Bank Marketing*, 36(2), 230-255. <https://doi.org/10.1108/ijbm-11-2016-0181>
- Md Morshedul Islam, A. A. M., amp, & Abu Saleh Muhammad, S. (2024). Enhancing Textile Quality Control With IOT Sensors: A Case Study Of Automated Defect Detection. *International Journal of Management Information Systems and Data Science*, 1(1), 19-30. <https://doi.org/10.62304/ijmisds.v1i1.113>
- Md Samiul Alam, M. (2024). The Transformative Impact of Big Data in Healthcare: Improving Outcomes, Safety, and Efficiencies. *Global Mainstream Journal of Business, Economics, Development & Project Management*, 3(03), 01-12. <https://doi.org/10.62304/jbedpm.v3i03.82>
- Mintoo, A. A. (2024a). Data-Driven Journalism: Advancing News Reporting Through Analytics With A PRISMA-Guided Review. *Journal of Machine Learning, Data Engineering and Data Science*, 1(01), 19-40. <https://doi.org/10.70008/jmldeds.v1i01.39>
- Mintoo, A. A. (2024b). Detecting Fake News Using Data Analytics: A Systematic Literature Review And Machine Learning Approach. *Academic Journal on Innovation, Engineering & Emerging Technology*, 1(01), 108-130. <https://doi.org/10.69593/ajieet.v1i01.143>
- Monferrer-Tirado, D., Estrada-Guillén, M., Fandos-Roig, J. C., Moliner-Tena, M. Á., & García, J. S. (2016). Service quality in bank during an economic crisis. *International Journal of Bank Marketing*, 34(2), 235-259. <https://doi.org/10.1108/ijbm-01-2015-0013>
- Morosan, C. (2010). Theoretical and Empirical Considerations of Guests' Perceptions of Biometric Systems in Hotels: Extending the Technology Acceptance Model. *Journal of Hospitality & Tourism Research*, 36(1), 52-84. <https://doi.org/10.1177/1096348010380601>
- Mosleuzzaman, M., Hussain, M. D., Shamsuzzaman, H. M., Mia, A., & Hossain, M. D. S. (2024). Electric Vehicle Powertrain Design: Innovations In Electrical Engineering. *Academic Journal on Innovation, Engineering & Emerging Technology*, 1(01), 1-18. <https://doi.org/10.69593/ajieet.v1i01.114>
- Page, C. M., & Lepkowska-White, E. (2002). Web equity: a framework for building consumer value in online companies. *Journal of Consumer Marketing*, 19(3), 231-248. <https://doi.org/10.1108/07363760210426058>
- Pons, A. P., & Polak, P. (2008). Understanding user perspectives on biometric technology. *Communications of the ACM*, 51(9), 115-118. <https://doi.org/10.1145/1378727.1389971>
- Rahaman, M. A., Rozony, F. Z., Mazumder, M. S. A., & Haque, M. N. (2024). Big Data-Driven Decision Making in Project Management: A

- Comparative Analysis. *Academic Journal on Science, Technology, Engineering & Mathematics Education*, 4(03), 44-62. <https://doi.org/10.69593/ajsteme.v4i03.88>
- Saaty, T. L. (1988). *What is the analytic hierarchy process* (Vol. NA). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-83555-1_5
- Safeena, R., Date, H., Kammani, A., & Hundewale, N. (2012). Technology Adoption and Indian Consumers: Study on Mobile Banking. *International Journal of Computer Theory and Engineering*, NA(NA), 1020-1024. <https://doi.org/10.7763/ijcte.2012.v4.630>
- Sebayang, T. E., Hakim, D. B., Bakhtiar, T., & Indrawan, D. (2023). What Accelerates the Choice of Mobile Banking for Digital Banks in Indonesia? *Journal of Risk and Financial Management*, 17(1), 6-6. <https://doi.org/10.3390/jrfm17010006>
- Sebayang, T. E., Hakim, D. B., Bakhtiar, T., & Indrawan, D. (2024). The Investigation of Preference Attributes of Indonesian Mobile Banking Users to Develop a Strategy for Mobile Banking Adoption. *Journal of Risk and Financial Management*, 17(3), 109-109. <https://doi.org/10.3390/jrfm17030109>
- Shaikh, A. A., & Karjaluo, H. (2015). Mobile banking adoption: A literature review. *Telematics and Informatics*, 32(1), 129-142. <https://doi.org/10.1016/j.tele.2014.05.003>
- Shamim, M. (2022). The Digital Leadership on Project Management in the Emerging Digital Era. *Global Mainstream Journal of Business, Economics, Development & Project Management*, 1(1), 1-14
- Shamsuzzaman, H. M., Mosleuzzaman, M. D., Mia, A., & Nandi, A. (2024). Cybersecurity Risk Mitigation in Industrial Control Systems Analyzing Physical Hybrid And Virtual Test Bed Applications. *Academic Journal on Artificial Intelligence, Machine Learning, Data Science and Management Information Systems*, 1(01), 19-39. <https://doi.org/10.69593/ajaimldsmis.v1i01.123>
- Shorna, S. A., Sultana, R., & Hasan, Molla Al R. (2024a). Big Data Applications in Remote Patient Monitoring and Telemedicine Services: A Review of Techniques and Tools. *Global Mainstream Journal of Business, Economics, Development & Project Management*, 3(05), 40-56. <https://doi.org/10.62304/jbedpm.v3i05.206>
- Shorna, S. A., Sultana, R., & Hasan, M. A. R. (2024b). Transforming Healthcare Delivery Through Big Data in Hospital Management Systems: A Review of Recent Literature Trends. *Academic Journal on Artificial Intelligence, Machine Learning, Data Science and Management Information Systems*, 1(01), 1-18. <https://doi.org/10.69593/ajaimldsmis.v1i01.117>
- Sohel, A., Alam, M. A., Waliullah, M., Siddiki, A., & Uddin, M. M. (2024). Fraud Detection in Financial Transactions Through Data Science For Real-Time Monitoring And Prevention. *Academic Journal on Innovation, Engineering & Emerging Technology*, 1(01), 91-107. <https://doi.org/10.69593/ajieet.v1i01.132>
- Sultana, R., & Aktar, M. N. (2024). Artificial Intelligence And Big Data For Enhancing Public Health Surveillance And Disease Prevention: A Systematic Review. *Journal of Machine Learning, Data Engineering and Data Science*, 1(01), 129-146. <https://doi.org/10.70008/jmldeds.v1i01.50>
- Tallon, P. P. (2010). A Service Science Perspective on Strategic Choice, IT, and Performance in U.S. Banking. *Journal of Management Information Systems*, 26(4), 219-252. <https://doi.org/10.2753/mis0742-1222260408>
- Taylor, P. J., Dargahi, T., Dehghantanha, A., Parizi, R. M., & Choo, K.-K. R. (2020). A systematic literature review of blockchain cyber security. *Digital Communications and Networks*, 6(2), 147-156. <https://doi.org/10.1016/j.dcan.2019.01.005>
- Tinnilä, M. (2012). Impact of Future Trends on Banking Services. *The Journal of Internet Banking and Commerce*, 17(2), 1-15. <https://doi.org/NA>
- Trivedi, J. P. (2019). Examining the Customer Experience of Using Banking Chatbots and Its Impact on Brand Love: The Moderating Role of Perceived Risk. *Journal of Internet Commerce*, 18(1), 91-111. <https://doi.org/10.1080/15332861.2019.1567188>
- Venkatesh, V., Morris, M. G., Davis, G. B., & Davis, F. D. (2003). User acceptance of information technology: toward a unified view. *MIS Quarterly*, 27(3), 425-478. <https://doi.org/10.2307/30036540>

- Wang, S. W., Ngamsiriudom, W., & Hsieh, C.-H. (2015). Trust disposition, trust antecedents, trust, and behavioral intention. *The Service Industries Journal*, 35(10), 555-572. <https://doi.org/10.1080/02642069.2015.1047827>
- Yadav, M., & Rahman, Z. (2017). Social media marketing: literature review and future research directions. *International Journal of Business Information Systems*, 25(2), 213-213. <https://doi.org/10.1504/ijbis.2017.083687>
- Yousafzai, S., Foxall, G. R., & Pallister, J. G. (2010). Explaining Internet Banking Behavior: Theory of Reasoned Action, Theory of Planned Behavior, or Technology Acceptance Model? *Journal of Applied Social Psychology*, 40(5), 1172-1202. <https://doi.org/10.1111/j.1559-1816.2010.00615.x>
- Zeithaml, V. A., Parasuraman, A., & Malhotra, A. (2002). Service Quality Delivery Through Web Sites: A Critical Review of Extant Knowledge. *Journal of the Academy of Marketing Science*, 30(4), 362-375. <https://doi.org/10.1177/009207002236911>
- Zhou, T. (2012). Understanding users' initial trust in mobile banking: An elaboration likelihood perspective. *Computers in Human Behavior*, 28(4), 1518-1525. <https://doi.org/10.1016/j.chb.2012.03.021>